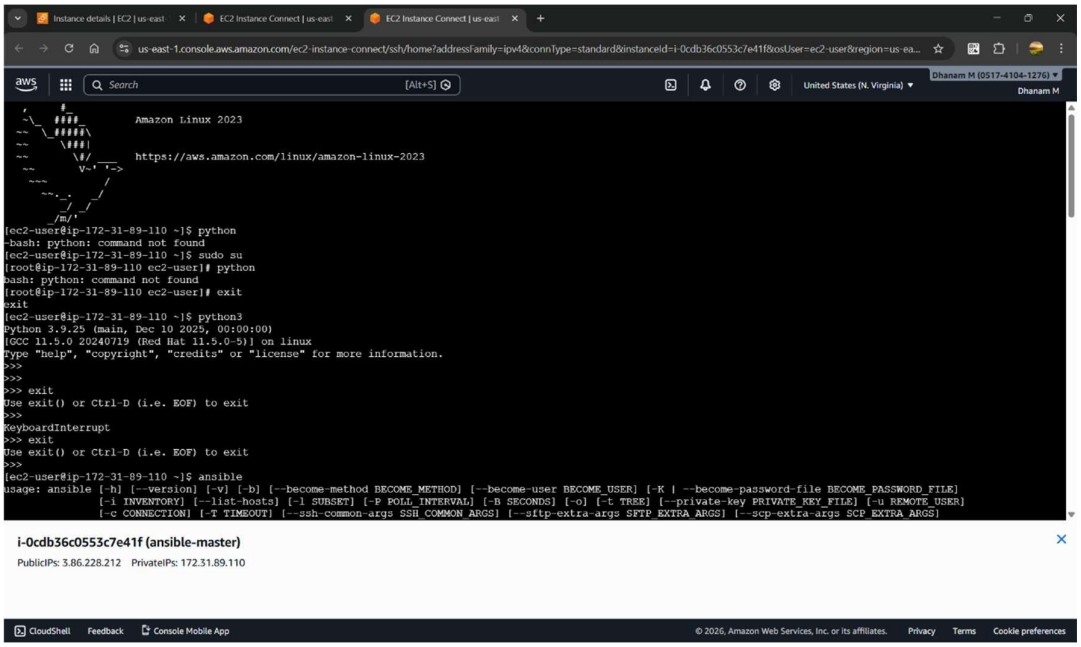
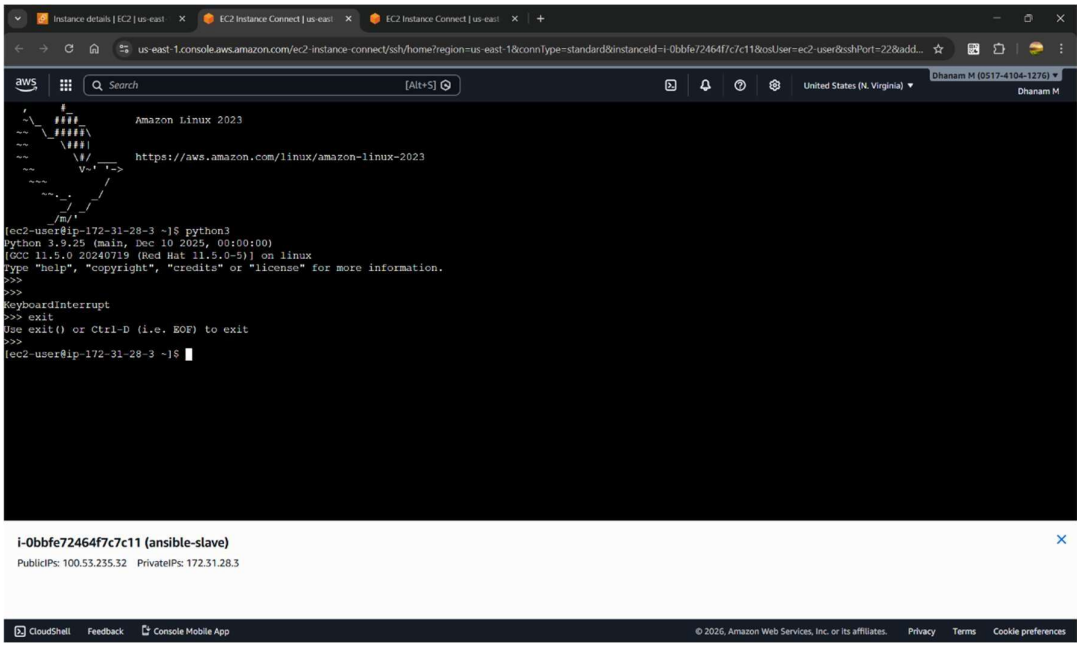
WEEKLY TEST

ANSIBLE MASTER AND SLAVE

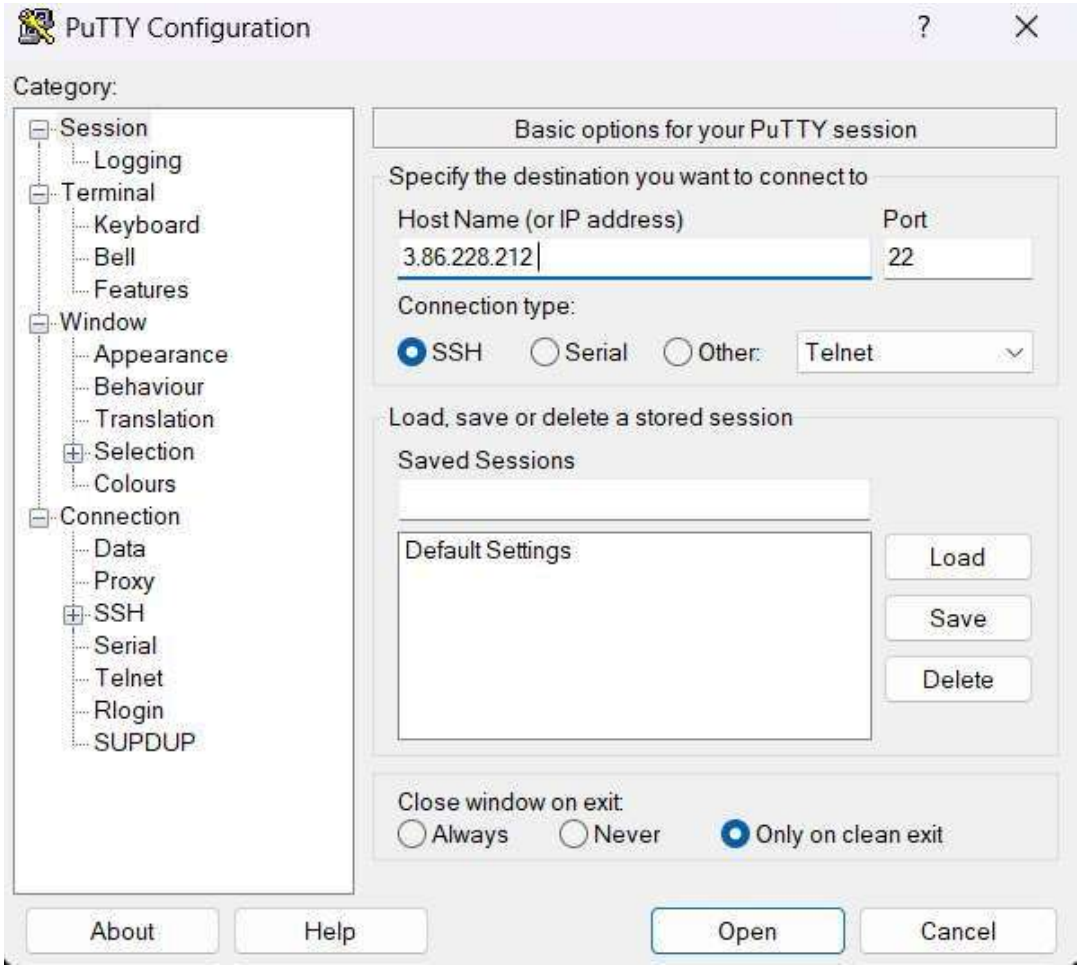
- ansible master to install an python3 -pip and install an ansible using an amazon linux machine.



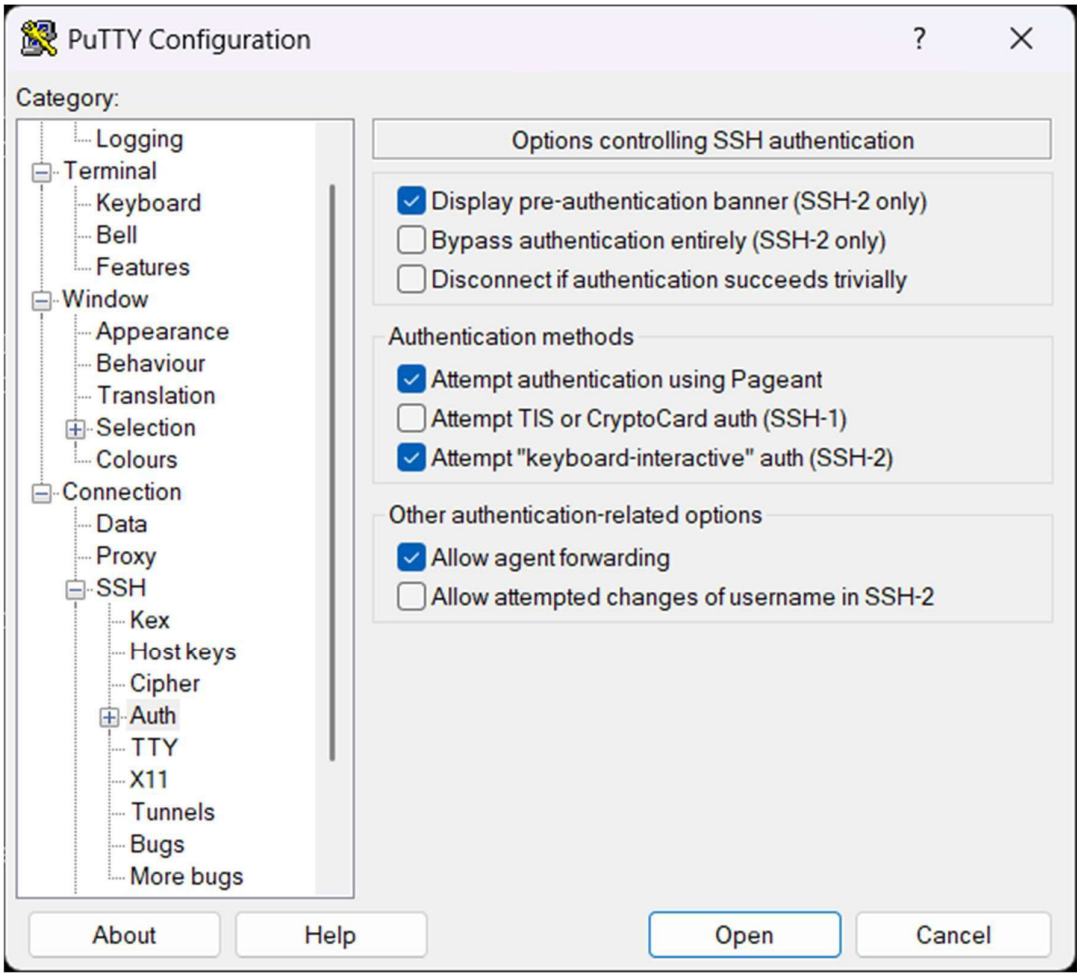
- ansible slave to install python3-pip using an amazon linux.



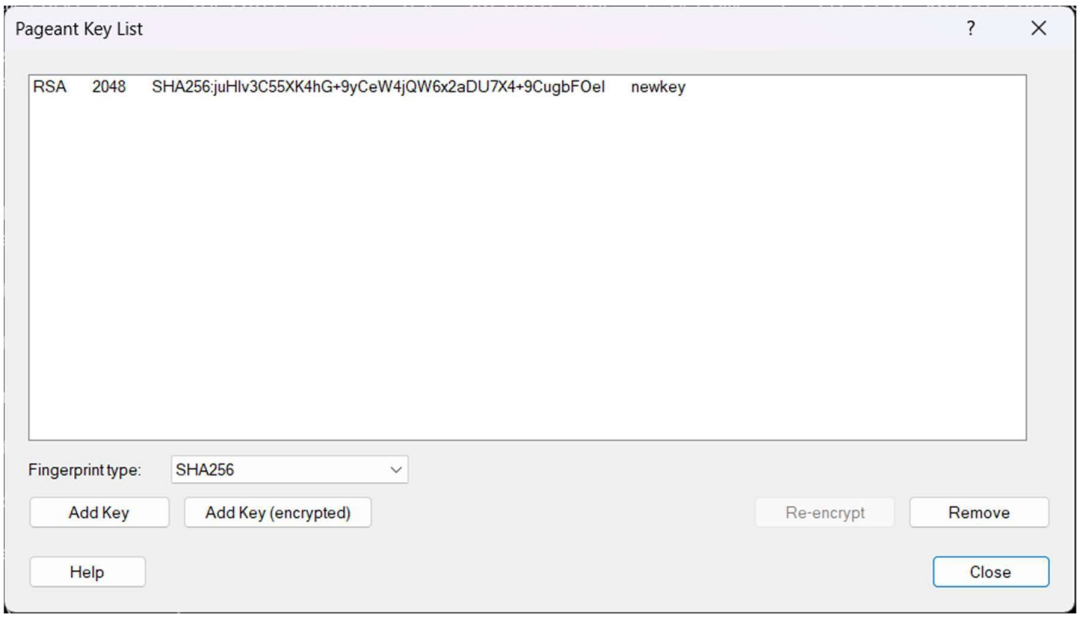
- to download an putty.exe and open an putty to copy master public ip to paste host in putty.



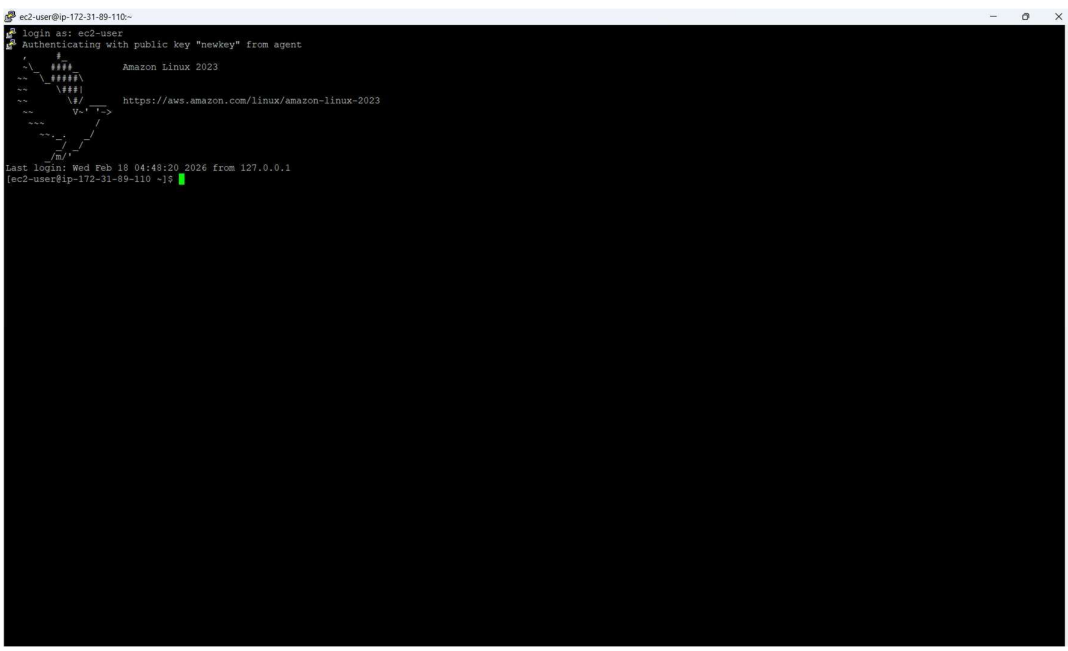
- to click an ssh and open to click authorized and allow agent forwarding and click open.



- to open pagent.ex to add key ppk.key and close.



- login ec2-user.



- install httpd using ansible machine.

```
[ec2-user@ip-172-31-89-110 ~]$ ansible --version
ansible [core 2.15.3]
  config file = /home/ec2-user/ansible.cfg
  configured module search path = ['/home/ec2-user/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  ansible collection location = /home/ec2-user/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.9.25 (main, Dec 10 2025, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3.9)
  jinja version = 3.1.4
  libyaml = True
[ec2-user@ip-172-31-89-110 ~]$ ansible localhost -m yum -a "name=httpd state=present" -b -i localhost,
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning
of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
localhost | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "msg": "",
    "rc": 0,
    "results": [
        "Installed: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch",
        "Installed: httpd-core-2.4.66-1.amzn2023.0.1.x86_64",
        "Installed: mailcap-2.1.49-3.amzn2023.0.3.noarch",
        "Installed: apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64",
```

```
ec2-user@ip-172-31-89-110:~
    "changed": false,
    "ping": "pong"
}
[ec2-user@ip-172-31-89-110 ~]$ ansible --version
ansible [core 2.15.3]
  config file = /home/ec2-user/ansible.cfg
  configured module search path = ['/home/ec2-user/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  ansible collection location = /home/ec2-user/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.9.25 (main, Dec 10 2025, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3.9)
  jinja version = 3.1.4
  libyaml = True
[ec2-user@ip-172-31-89-110 ~]$ ansible localhost -m yum -a "name=httpd state=present" -b -i localhost,
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning
of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
localhost | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "msg": "",
    "rc": 0,
    "results": [
        "Installed: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch",
        "Installed: httpd-core-2.4.66-1.amzn2023.0.1.x86_64",
        "Installed: mailcap-2.1.49-3.amzn2023.0.3.noarch",
        "Installed: apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64",
        "Installed: mod_lua-2.4.66-1.amzn2023.0.1.x86_64",
        "Installed: apr-1.7.5-1.amzn2023.0.4.x86_64",
        "Installed: httpd-tools-2.4.66-1.amzn2023.0.1.x86_64",
        "Installed: apr-util-1.6.3-1.amzn2023.0.2.x86_64",
        "Installed: mod_http2-2.0.27-1.amzn2023.0.3.x86_64",
        "Installed: httpd-2.4.66-1.amzn2023.0.1.x86_64",
        "Installed: generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch",
        "Installed: apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64",
        "Installed: libbrotli-1.0.9-4.amzn2023.0.2.x86_64"
    ]
}
[ec2-user@ip-172-31-89-110 ~]$ ansible localhost -m service -a "name=httpd state=started enabled=yes" -b -i localhost,
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning
of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
localhost | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "enabled": true,
    "name": "httpd",
    "state": "started",
    "status": {
        "AccessSELinuxContext": "system_u:object_r:httpd_unit_file_t:s0",
        "ActiveEnterTimestampMonotonic": "0",
        "ActiveExitTimestampMonotonic": "0",
        "ActiveState": "inactive",
```

It works!

- run the host <public ip>:httpd port number 80.

Ansible - adhoc commands and playbook

What are Ad-Hoc Commands in Ansible?

An ad-hoc command is a one-line instruction used to perform a quick task on remote machines without writing a playbook.

It is imperative automation, not structured automation.

These are the example of ad hoc commands.

```
[ec2-user@ip-172-31-89-110 ~]$ ansible all -i inventory.txt -m copy -a "src=hari dest=/tmp"
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.27.165 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "checksum": "42ac8fad543cfa32fe27fea08a62beb5b9e299a8",
    "dest": "/tmp/hari",
    "gid": 1000,
    "group": "ec2-user",
    "md5sum": "fc93cab59237d57daf8cd657a540fbb1",
    "mode": "0664",
    "owner": "ec2-user",
    "secontext": "unconfined_u:object_r:user_home_t:s0",
    "size": 13,
    "src": "/home/ec2-user/.ansible/tmp/ansible-tmp-1771433383.2450376-3612-252473401499282/source",
    "state": "file",
    "uid": 1000
}
[WARNING]: Platform linux on host 172.31.28.3 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.28.3 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "checksum": "42ac8fad543cfa32fe27fea08a62beb5b9e299a8",
    "dest": "/tmp/hari",
    "gid": 1000,
    "group": "ec2-user",
    "md5sum": "fc93cab59237d57daf8cd657a540fbb1",
    "mode": "0664",
    "owner": "ec2-user",
    "secontext": "unconfined_u:object_r:user_home_t:s0",
    "size": 13,
    "src": "/home/ec2-user/.ansible/tmp/ansible-tmp-1771433383.2159321-3611-163560848913361/source",
    "state": "file",
    "uid": 1000
}
[ec2-user@ip-172-31-89-110 ~]$
```

```
Instances | EC2 | us-east-1
EC2 Instance Connect | us-east
Ansible - Ad hoc Commands -
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0bbfe72464f7c7c11&osUser=ec2-user&region=us-eas...
aws  Search [Alt+S]  United States (N. Virginia)  Dhanam M (0517-4104-1276)  Dhanam M

       ,     #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
Last login: Wed Feb 18 16:49:44 2026 from 172.31.89.110
[ec2-user@ip-172-31-28-3 ~]$ ls
[ec2-user@ip-172-31-28-3 ~]$ cd/tmp
-bash: cd/tmp: No such file or directory
[ec2-user@ip-172-31-28-3 ~]$ cd /tmp
[ec2-user@ip-172-31-28-3 tmp]$ ls
hari                                                                    systemd-private-db2a7c08772d42a381822a2943f302fd-policy-routes@ens5.service-8b52df
systemd-private-db2a7c08772d42a381822a2943f302fd-chronyd.service-CmVi1TE systemd-private-db2a7c08772d42a381822a2943f302fd-systemd-logind.service-eslAuf
systemd-private-db2a7c08772d42a381822a2943f302fd-dbus-broker.service-Mj295G systemd-private-db2a7c08772d42a381822a2943f302fd-systemd-resolved.service-NGfy92
[ec2-user@ip-172-31-28-3 tmp]$

i-0bbfe72464f7c7c11 (ansible-slave)
PublicIPs: 52.73.204.225  PrivateIPs: 172.31.28.3
CloudShell  Feedback  Console Mobile App  © 2026, Amazon Web Services, Inc. or its affiliates.  Privacy  Terms  Cookie preferences
```

```
[ec2-user@ip-172-31-89-110 ~]$ ansible all -i inventory.txt -m command -a "uptime"
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.27.165 | CHANGED | rc=0 >>
 16:45:29 up 16 min,  1 user,  load average: 0.08, 0.03, 0.01
[WARNING]: Platform linux on host 172.31.28.3 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.28.3 | CHANGED | rc=0 >>
 16:45:29 up 16 min,  1 user,  load average: 0.00, 0.00, 0.00
```

```
[ec2-user@ip-172-31-89-110 ~]$ ansible all -i inventory.txt -m ping
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.27.165 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Platform linux on host 172.31.28.3 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.28.3 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[ec2-user@ip-172-31-89-110 ~]$ ansible all -i inventory.txt -m setup
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.27.165 | SUCCESS => {
    "ansible_facts": {
        "ansible_all_ipv4_addresses": [
            "172.31.27.165"
        ],
        "ansible_all_ipv6_addresses": [
            "fe80::8ff:e4ff:fe76:2a09"
        ],
        "ansible_apparmor": {
            "status": "disabled"
        },
        "ansible_architecture": "x86_64",
        "ansible_bios_date": "10/16/2017",
        "ansible_bios_vendor": "Amazon EC2",
        "ansible_bios_version": "1.0",
        "ansible_board_asset_tag": "i-0bb8079c314c6de2e",
        "ansible_board_name": "NA",
        "ansible_board_serial": "NA",
```

```
[ec2-user@ip-172-31-89-110 ~]$ ansible all -i inventory.txt -m file -a "path=/tmp/testdir mode=0755 state=directory"
[WARNING]: Platform linux on host 172.31.28.3 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.28.3 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "gid": 1000,
    "group": "ec2-user",
    "mode": "0755",
    "owner": "ec2-user",
    "path": "/tmp/testdir",
    "secontext": "unconfined_u:object_r:user_tmp_t:s0",
    "size": 40,
    "state": "directory",
    "uid": 1000
}
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.27.165 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "gid": 1000,
    "group": "ec2-user",
    "mode": "0755",
    "owner": "ec2-user",
    "path": "/tmp/testdir",
    "secontext": "unconfined_u:object_r:user_tmp_t:s0",
    "size": 40,
    "state": "directory",
    "uid": 1000
}
[ec2-user@ip-172-31-89-110 ~]$
```

```
[ec2-user@ip-172-31-28-3 tmp]$ ls
hari
systemd-private-db2a7c08772d42a381822a2943f302fd-chronyd.service-CmViTE
systemd-private-db2a7c08772d42a381822a2943f302fd-dbus-broker.service-Mj295G
systemd-private-db2a7c08772d42a381822a2943f302fd-policy-routes@ens5.service-8b52df
systemd-private-db2a7c08772d42a381822a2943f302fd-systemd-logind.service-eslAuf
systemd-private-db2a7c08772d42a381822a2943f302fd-systemd-resolved.service-NGfy92
testdir
[ec2-user@ip-172-31-28-3 tmp]$
```

```
i-0bbfe72464f7c7c11 (ansible-slave)
PublicIPs: 52.73.204.225   PrivateIPs: 172.31.28.3

CloudShell   Feedback   Console Mobile App   © 2026, Amazon Web Services, Inc. or its affiliates.   Privacy   Terms   Cookie preferences
```

What Is a Playbook?

A playbook in Ansible is a YAML file that defines automation in a structured, repeatable way.

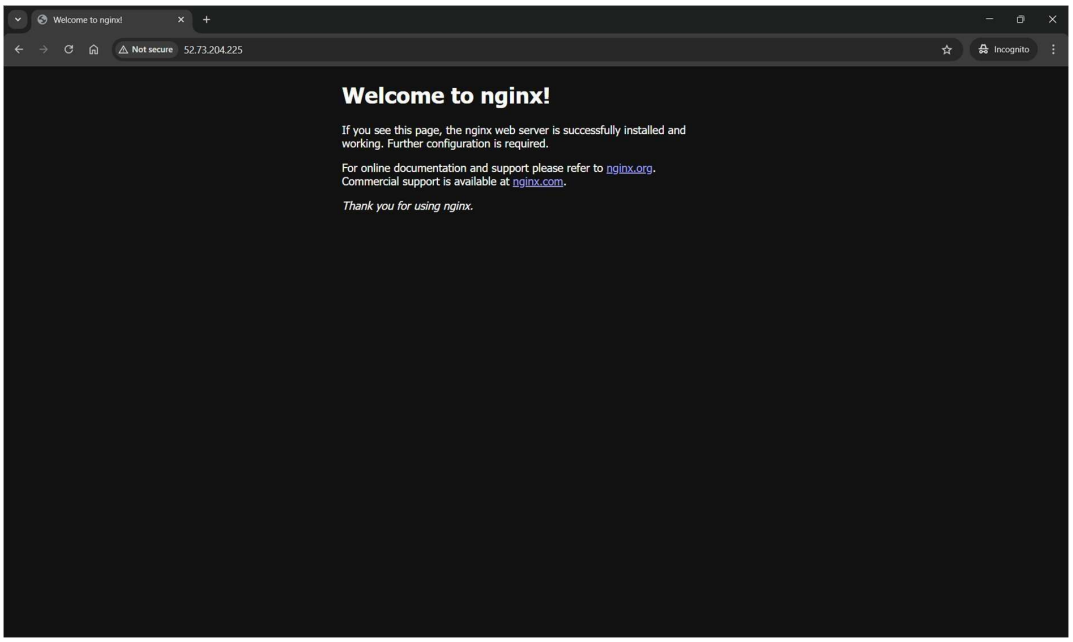
It describes:

- Which machines to target
- What tasks to run

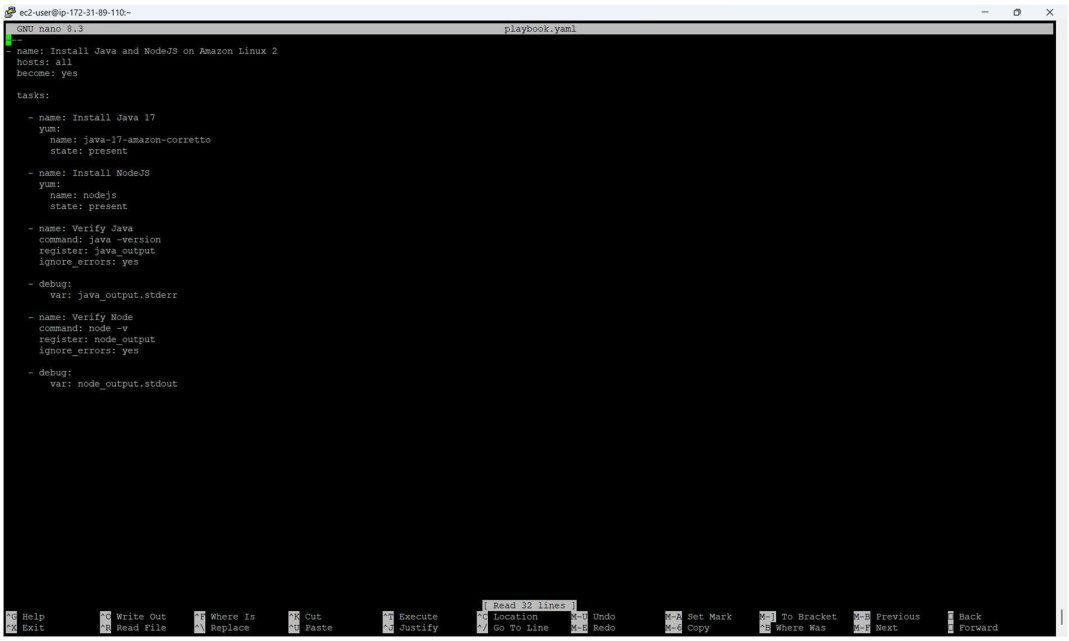
- In what order
- With what privileges
- Under what conditions

```
---
- name: Install and Start Nginx   hosts: webservers   become:
  yes
  tasks:
- name: Install nginx
      apt:
        name: nginx
state: present     - name:
Start nginx       service:
name: nginx         state:
started
```

I have installed the nginx and hosted on the web.



And then I have installed the JAVA , NODE.js .



The process have been done using the playbook.

```
ec2-user@ip-172-31-89-110:~
      ignore_errors: yes

    - debug:
        var: node_output.stdout
[ec2-user@ip-172-31-89-110 ~]$ ansible-playbook -i inventory.txt playbook.yaml

PLAY [Install Java and NodeJS on Amazon Linux 2] ********************************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************************************************************
[WARNING]: Platform linux on host 172.31.28.3 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.28.3]
[WARNING]: Platform linux on host 172.31.27.165 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the
meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.27.165]

TASK [Install Java 17] ***********************************************************************************************************************************************
changed: [172.31.27.165]
changed: [172.31.28.3]

TASK [Install NodeJS] ************************************************************************************************************************************************
changed: [172.31.27.165]
changed: [172.31.28.3]

TASK [Verify Java] ***************************************************************************************************************************************************
changed: [172.31.27.165]
changed: [172.31.28.3]

TASK [debug] *********************************************************************************************************************************************************
ok: [172.31.28.3] => {
    "java_output.stderr": "openjdk version \"17.0.18\" 2026-01-20 LTS\nOpenJDK Runtime Environment Corretto-17.0.18.9.1 (build 17.0.18+9-LTS)\nOpenJDK 64-Bit Server VM Corretto-17.0.18.9.1
(build 17.0.18+9-LTS, mixed mode, sharing)"
}
ok: [172.31.27.165] => {
    "java_output.stderr": "openjdk version \"17.0.18\" 2026-01-20 LTS\nOpenJDK Runtime Environment Corretto-17.0.18.9.1 (build 17.0.18+9-LTS)\nOpenJDK 64-Bit Server VM Corretto-17.0.18.9.1
(build 17.0.18+9-LTS, mixed mode, sharing)"
}

TASK [Verify Node] ***************************************************************************************************************************************************
changed: [172.31.27.165]
changed: [172.31.28.3]

TASK [debug] *********************************************************************************************************************************************************
ok: [172.31.28.3] => {
    "node_output.stdout": "v18.20.8"
}
ok: [172.31.27.165] => {
    "node_output.stdout": "v18.20.8"
}

PLAY RECAP ***********************************************************************************************************************************************************
172.31.27.165              : ok=7    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.28.3                : ok=7    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

[ec2-user@ip-172-31-89-110 ~]$
```

Ansible – Roles.

A role in Ansible is a structured way to organize playbooks, variables, files, templates, and tasks into a reusable unit.

Why Roles Exist

Without roles:

- One huge YAML file
- Repeated code
- Hard to maintain
- Impossible to scale With roles:
- Modular structure
- Reusable across projects
- Clean separation of concerns
- Easier debugging

This is the example file for creating users in the slave machine.

```
ec2-user@ip-172-31-89-110:~
  GNU nano 8.3                    firstrole.yaml                    Modified
- name: Install multiple packages using loop
  hosts: all
  become: yes
  roles:
    - myroles

^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark  M-] To Bracket  M-B Previous  ^_ Back
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy      ^Q Where Was    M-F Next      ^_ Forward
```

This is the command to create an role.

```
[ec2-user@ip-172-31-91-97 ~]$ ansible-galaxy init myrole
- Role myrole was created successfully
```

The default directory of the role.

```
0 directories, 0 files
[ec2-user@ip-172-31-89-110 firstrole]$ ls
README.md  defaults  files  handlers  meta  tasks  templates  tests  vars
[ec2-user@ip-172-31-89-110 firstrole]$ tree
.
├── README.md
├── defaults
│   └── main.yml
├── files
├── handlers
│   └── main.yml
├── meta
│   └── main.yml
├── tasks
│   └── main.yml
├── templates
├── tests
│   ├── inventory
│   └── test.yml
└── vars
    └── main.yml

8 directories, 8 files
[ec2-user@ip-172-31-89-110 firstrole]$
```

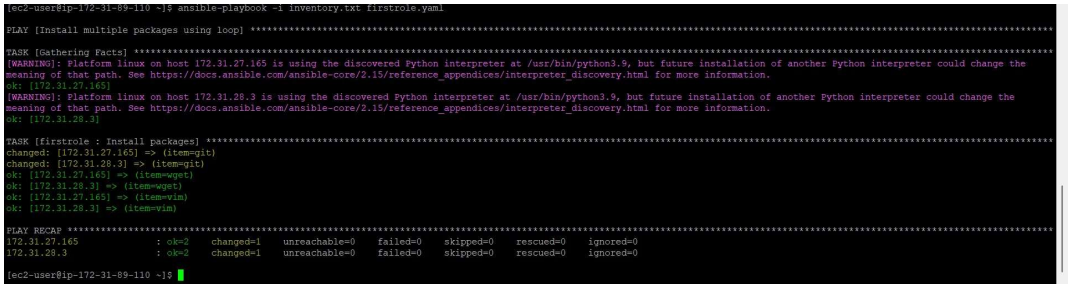
We have to create an role file in the tasks to put the tasks in the directory.

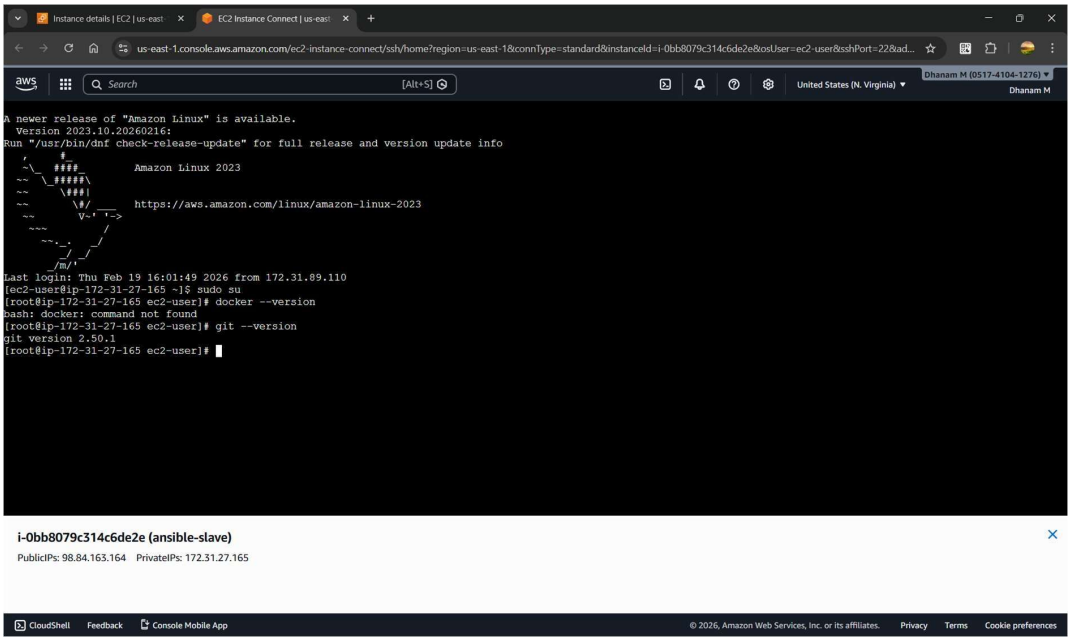
```
[ec2-user@ip-172-31-91-97 ~]$ cd myrole/
[ec2-user@ip-172-31-91-97 myrole]$ cd tas
-bash: cd: tas: No such file or directory
[ec2-user@ip-172-31-91-97 myrole]$ cd tasks/
[ec2-user@ip-172-31-91-97 tasks]$ nano main.yml
```

```
ec2-user@ip-172-31-89-110:~/firstrole/tasks
  GNU nano 8.3                                  main.yml
---
# tasks file for firstrole
  - name: Install packages
    yum:
      name: "{{ item }}"
      state: present
    loop:
      - git
      - wget
      - vim

                                   [ Read 10 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark   M-] To Bracket  M-Q Previous   ^B Back
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^Q Where Was    M-W Next       ^F Forward
```

Then we have to use the ansible command to call the file.





The process have been executed successfully.